

2(a)	1 cm ³ of stock solution to 149 cm ³ of distilled water ; (dilution factor is) 150 ;	2
2(b)	<i>any one from:</i> 1 measure from, top of the soil / bottom of stem / AW, to top of the stem / AW ; 2 straighten the stem (while measuring length) ; 3 place string along stem, and measure length of string ;	1
2(c)(i)	1.9 ;	1
2(c)(ii)	<i>any one from:</i> 1 (mean stem) length was measured, in / to the nearest, cm ; 2 many anomalies in the results / identified anomaly ; 3 no statistical test carried out ;	1
2(c)(iii)	<i>any one from:</i> 1 measure lengths in mm ; 2 <i>idea of</i> repeat high concentration GA ₃ experiment (to check anomaly) ; 3 measure, at same time every day / for more days ; 4 measure more seedlings / increase sample size ;	1
2(d)	(SE =) 0.067 ; (95% CI =) 1.38 ± 0.13 ;	2

Question	Answer	Marks
2(e)	any three from: 1 highest (concentration of) GA₈ with, red light / 680 nm or lowest (concentration of) GA₁ with, blue light / 470 nm or highest (concentration of) GA₁, in the dark / with no light ; 2 lower (concentration of) GA₁ for all wavelengths of light (compared to dark) or higher (concentration of) GA₈ for all wavelengths of light (compared to dark) ; 3 95% CI, does not overlap (for the lower concentration of GA₁ for all wavelengths of light), so there is a significant difference or 95% CI, does not overlap, (higher concentration of GA₈ for all of the wavelengths of light), so there is a significant difference ;	3